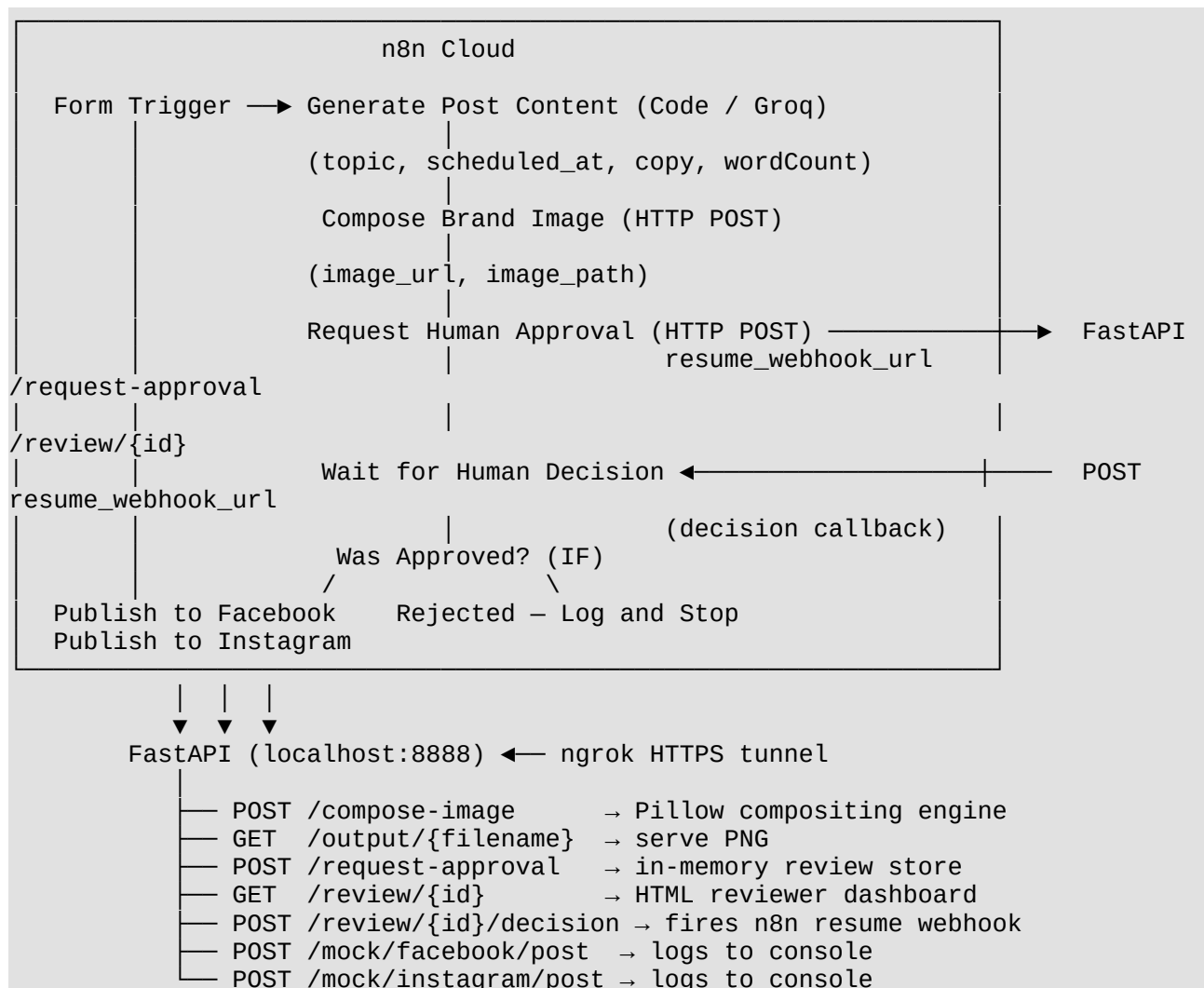


System Architecture

Component map



Data flow — step by step

1. Form submission

User submits the n8n Form Trigger with:

- `topic` — free-text post subject
- `scheduled_at` — ISO 8601 publish time

2. LLM content generation (n8n Code node)

- Calls Groq (Llama 3.3 70B) with a `system` message enforcing 90–150 words
- Counts words after each response; sends corrective multi-turn prompts on failure

- If still out of range after 5 attempts, the node throws — strict guarantee that no out-of-range post proceeds
- Outputs: { topic, scheduled_at, copy, wordCount, attempts, withinRange }

3. Image compositing (n8n HTTP Request → FastAPI)

- n8n POSTs { title: topic } to POST /compose-image
- FastAPI runs `compositing.py`: opens template PNG, pastes logo at fixed pixel coords, wraps and draws title text in the lower third
- Saves PNG to `backend/output/`, returns { image_url, image_path }

4. HITL request (n8n HTTP Request → FastAPI)

- n8n POSTs { copy, image_url, resume_webhook_url } to POST /request-approval
 - `resume_webhook_url = $execution.resumeUrl` — n8n's unique per-execution webhook
- FastAPI stores the review in `_reviews[review_id]` and returns { review_id, review_url }

5. Workflow suspension (n8n Wait node)

- n8n serialises execution state and suspends — zero CPU usage
- Execution can wait indefinitely (or until a configured timeout)

6. Human review

- Reviewer opens `http://NGROK_URL/review/{review_id}` in a browser
- Dashboard shows: composited image, generated copy in an editable textarea, three buttons
- Reviewer clicks **Approve**, **Edit & Approve** (after modifying the copy), or **Reject**

7. Decision callback (FastAPI → n8n)

- FastAPI POST `/review/{id}/decision` fires `requests.post(resume_webhook_url, json={ decision, final_copy, review_id })`
- n8n receives the POST and resumes the paused execution
- The Wait node nests the POST payload under `$json.body`, so the Was Approved? IF node checks `$json.body.decision`

8. Publish or stop

- **Approved:** n8n POSTs to `/mock/facebook/post` then `/mock/instagram/post`; FastAPI logs the payload to the terminal

- **Rejected:** n8n runs the Set node, sets `status = "rejected"`, execution ends

Why each boundary exists

Boundary	Reason
n8n ↔ FastAPI over HTTP	n8n has no Python runtime; FastAPI gives pixel-level Pillow control
ngrok tunnel	n8n Cloud (hosted) cannot reach localhost directly
In-memory review store	Sufficient for demo scope; trivially replaceable with Redis/Postgres
<code>\$execution.resumeUrl</code> pattern	The correct n8n primitive for async human-in-the-loop — no polling, no external state machine
Groq free tier	Zero cost; very fast inference (sub-second), generous free rate limits

Sequence diagram — approve path

